

Fibroblasts extracellular vesicles modulate human keratinocytes differentiation during skin aging

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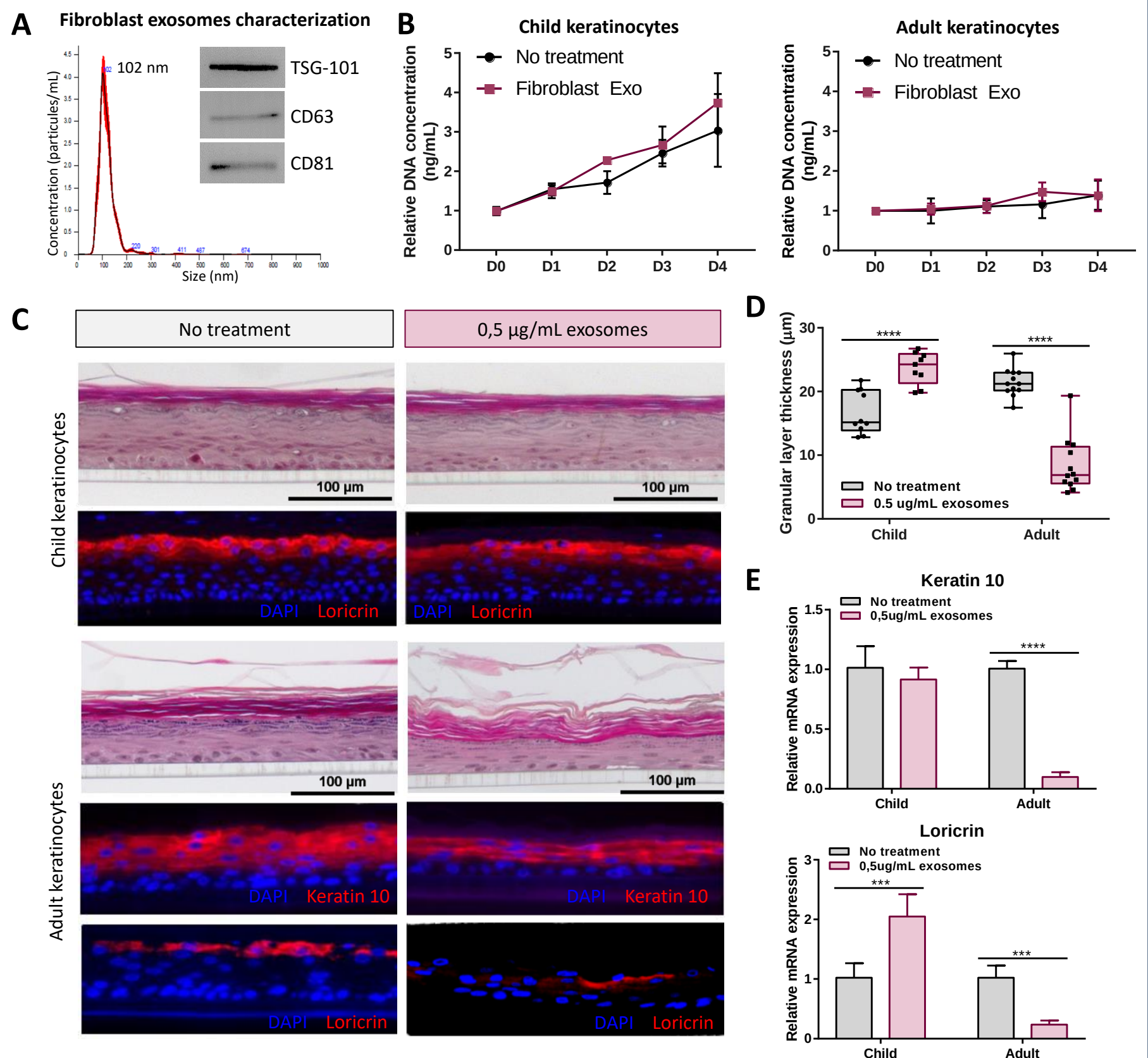
Oral presentation 052 (session 7)

Context

Small extracellular vesicles (EVs), ie, microvesicles and exosomes, once considered as mechanisms of waste elimination, are now emerging as key players in intercellular communications. EVs are made up of a lipid bilayer encapsulating proteins, messenger or non-coding RNAs as well as lipids which can be captured by recipient cells, thus modifying their behavior. EVs play different roles in skin functions and homeostasis. The balance between proliferation and differentiation is essential for epidermis homeostasis and is mostly altered during skin aging. In this context, we wonder if EVs, especially exosomes, from dermal primary fibroblasts could be implicated in this balance and if these interactions are modified during life using keratinocytes from infant (<10 years old) and adults (19-46 years old).

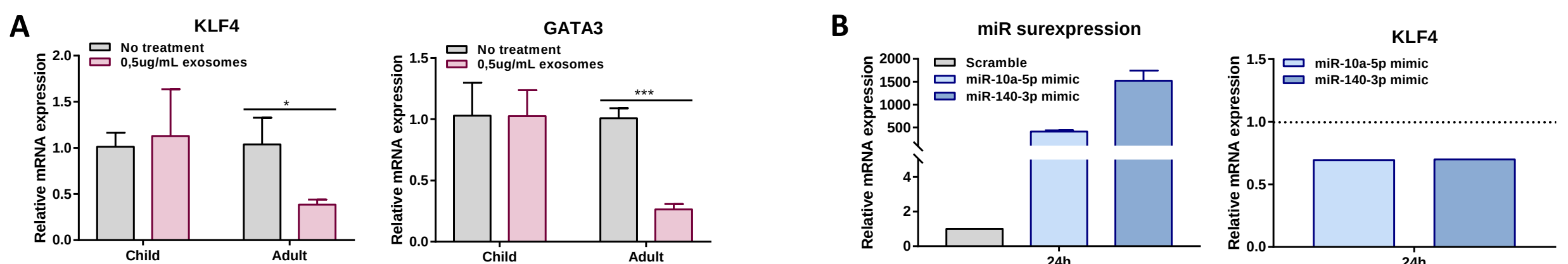
The effects of fibroblast exosomes on keratinocytes

- Size profile (NanoTracking Analysis) and protein markers of exosomes (Western blotting) in fibroblasts EVs.
- Fibroblasts exosomes do not modify proliferation of infant and adult keratinocytes.
- Fibroblast exosomes disrupt epidermis thickness and structure and modify protein expression of keratin 10 and loricrin in 3D reconstructed human epidermis model (RHE).
- According to the keratinocyte's age, fibroblast exosomes induce opposite effect on granular layer thickness in RHE.
- In RHE, fibroblast exosomes respectively induce increase and decrease of loricrin gene expression in child or adult primary keratinocytes. Keratin 10 gene expression is only decreased in adult condition.



Regulation of differentiation

- KLF4 and GATA3, two inducers of keratinocyte differentiation, are down-regulated in adult RHE treated with exosomes.
- Two microRNAs carried by fibroblasts exosomes regulate KLF4 expression : miR-10a-5p and miR-140-3p.



Conclusion

Fibroblasts exosomes differently modulate primary keratinocyte differentiation according to the age of donors: activation or repression in child and adult keratinocytes respectively. This repression could be linked to 2 microRNAs targeting KLF4, which is a transcription factor implicated in keratinocyte differentiation. In conclusion, these data show that EVs mediated intercellular communication is impacted by aging, through the intrinsic modifications of receiving cells during individual life.

